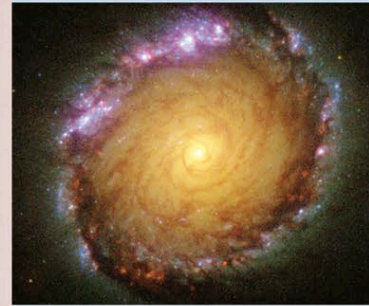


IRREGULAR DWARF

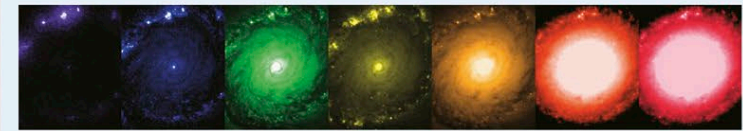
The irregular dwarf galaxy NGC 4449 contains clusters of young bluish stars interspersed with dustier reddish regions of current star formation.

EXPLORING SPACE

GALAXIES AT DIFFERENT WAVELENGTHS



COMBINED IMAGE OF NGC 1512



FAR ULTRA-VIOLET NEAR ULTRA-VIOLET GREEN VISIBLE LIGHT YELLOW VISIBLE LIGHT NEAR INFRARED MID-INFRARED FAR INFRARED

Radiation of different wavelengths can reveal hidden structures within galaxies. The hottest stars appear brightest in ultraviolet, while cool, diffuse gas may be visible only in infrared. By overlaying images from different spectral regions, astronomers build up a full picture of a galaxy.

FROM ULTRAVIOLET TO INFRARED

These images of galaxy NGC 1512 increase in wavelength from left to right. Each wavelength is represented by a false color.

IRREGULAR GALAXIES

Not all galaxies fit into the scheme of spirals, ellipticals, and lenticulars. Some of these misfit galaxies are colliding with companions or being pulled out of shape by a neighbor's gravity. These are usually classed under the catch-all term "peculiar" or "Pec." Many more are true irregulars (type Irr). These galaxies typically contain a lot of gas, dust, and hot blue stars. In fact, many irregulars are "starburst" galaxies, with great waves of star formation sweeping through them. Irregulars frequently have vast pink hydrogen-emission nebulae where star formation is taking place. Some irregulars show signs of structure—central bars and sometimes the beginnings of spiral arms. The Milky Way's brightest companion galaxies, the Large and Small Magellanic Clouds (see pp.310–311), are typical irregular galaxies.

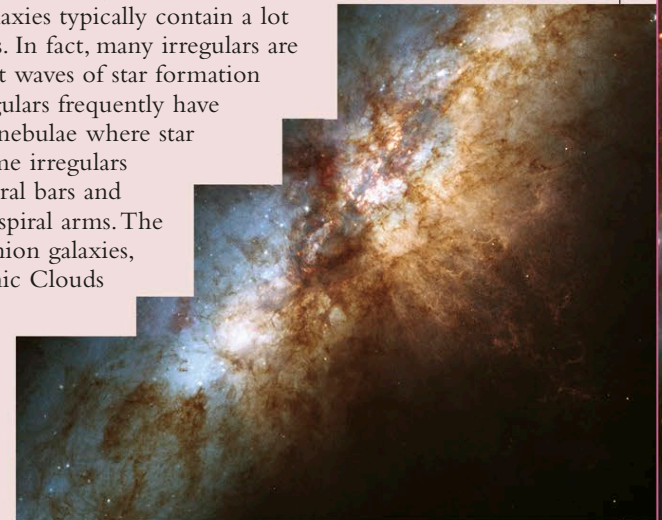
IRREGULAR STARBURST

M82 is an irregular starburst galaxy crossed with dark dust lanes. It is undergoing an intense period of star birth.



PECULIAR GALAXY

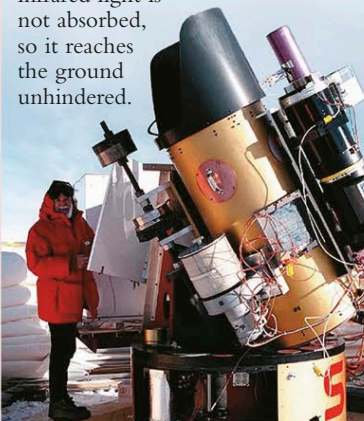
NGC 4650A is a rare example of a polar-ring galaxy, possibly created in a galactic collision. A blue-white star-forming ring, aligned with the poles, extends from the nucleus.



EXPLORING SPACE

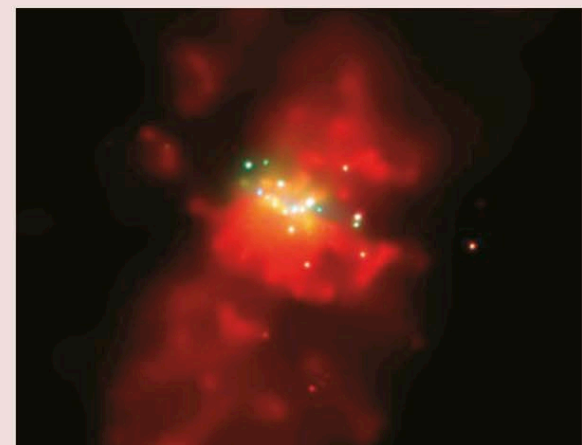
ASTRONOMY FROM THE SOUTH POLE

Some of the best Earth-based observations of galaxies come from an automated observatory at the South Pole. The AASTO project takes advantage of the dryness on the Antarctic Plateau—the driest place on Earth. With no water vapor in the atmosphere, near-infrared light is not absorbed, so it reaches the ground unhindered.



CENTRAL BLACK HOLES

Many, if not all, galaxies have a dark region within their nucleus that seems strange by contrast with the outer parts. The fast orbits of stars near galactic nuclei suggest an enormous concentration of mass in a tiny volume at the center of most spiral and elliptical galaxies—often billions of Suns' worth of material in a space little larger than the solar system. The only object that can reach such a density is a black hole (see p.26). Despite the tremendous gravity of this "supermassive" black hole, in most nearby galaxies the material has long since settled into steady orbits around it. With no material to absorb, the black hole remains dormant. When a gas cloud or other object comes too close, however, the black hole may awake, pulling in the stray material and heating it, producing radiation. The black hole may generate any type of radiation, from low-energy radio waves to high-energy X-rays. In extreme cases called "active galaxies" (see pp.320–321), the radiation from the nucleus is the galaxy's dominant feature.



HIDDEN SUPERMASSIVE BLACK HOLE

An X-ray image of galaxy M82 shows glowing hot gas and intense point sources of X-rays. These are probably stellar-mass black holes surrounding a central supermassive black hole.